

Empirical Evaluation of HermiT-Based Ontology Validation Across Multiple Knowledge Graph Datasets

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Abstract—Knowledge Graph (KG) validation is essential for identifying structural and semantic inconsistencies in graph-based knowledge representations. This study evaluates an ontology-based validation approach using the HermiT OWL 2 reasoner across five heterogeneous KG datasets: Wikidata, DBpedia, FB15K-237, WN18RR, and YAGO3-10. The evaluation focuses on four applicable error categories: Domain, Range, Duplicate Fact, and Missing Relation. A standardized validation pipeline was implemented using controlled samples and a common experimental configuration. The results demonstrate that HermiT-based validation provides strong performance for duplicate-fact detection, while performance varies across domain, range, and missing-relation validation depending on the structural characteristics of the datasets. The study also identifies applicability limitations arising from the semantic information available in each dataset. The results provide an empirical basis for assessing the applicability of ontology-based KG validation across heterogeneous datasets.

Index Terms—Knowledge Graph Validation, Ontology-Based Validation, HermiT, OWL 2, Knowledge Graphs, Semantic Validation

I. INTRODUCTION

Knowledge Graphs (KGs) represent entities and relationships in a graph structure and are widely used in applications such as semantic search, recommendation systems, question answering, and information integration. However, real-world KGs may contain erroneous, incomplete, duplicated, or semantically incompatible information.

KG validation techniques aim to identify such problems by applying structural, semantic, logical, or statistical constraints. Among these techniques, ontology-based validation uses formal semantic representations and automated reasoning to determine whether KG assertions satisfy predefined constraints.

This work evaluates an ontology-based validation approach using the HermiT OWL 2 reasoner across five heterogeneous datasets. The objective is not to propose a new validation algorithm, but to empirically evaluate the applicability and performance of HermiT-based validation across different datasets and error categories.

The evaluated datasets are Wikidata, DBpedia, FB15K-237, WN18RR, and YAGO3-10. Four error categories are evaluated where sufficient information is available: Domain Error, Range Error, Duplicate Fact, and Missing Relation.

II. OBJECTIVE

The primary objective of the experiment is to empirically evaluate the applicability of ontology-based validation using HermiT across multiple KG datasets.

The specific objectives are:

- Evaluate HermiT-based validation across five KG datasets.
- Measure validation performance for Domain, Range, Duplicate Fact, and Missing Relation errors.
- Maintain a common experimental configuration across datasets.
- Identify dataset-dependent applicability limitations.
- Provide reproducible experimental results for comparison with other KG validation categories.

III. DATASETS

Five KG datasets were evaluated. The datasets represent different structural and semantic characteristics.

TABLE I
DATASETS USED IN THE HERMIT EXPERIMENT

Dataset	Source Triples	Sample	Runtime (s)
Wikidata	64,163	2,000	7.070
DBpedia	89,794	2,000	11.575
FB15K-237	310,116	2,000	10.680
WN18RR	93,003	2,000	5.515
YAGO3-10	1,089,040	2,000	5.735

For Wikidata and DBpedia, the experiment used the available movie-domain KG files. FB15K-237, WN18RR, and YAGO3-10 were evaluated using their locally available dataset splits.

A controlled sample of up to 2,000 triples was used for the main experiment to maintain a common computational scale across datasets.

IV. EXPERIMENTAL ENVIRONMENT

The implementation was developed using Python and Owlready2 with the HermiT OWL 2 reasoner.

The main software configuration was:

- Python: 3.14.7
- Owlready2: 0.51
- HermiT: 1.3.8.1099
- Java: 21
- Random seed: 42

The same validation implementation and experimental configuration were applied across the five datasets to improve reproducibility and comparability.

V. ONTOLOGY CONSTRUCTION

For each dataset, an OWL ontology was constructed from the corresponding KG sample. Individuals were generated from KG entities, while relations were represented as OWL object properties.

Dataset-specific relation signatures were extracted from the available training information. These signatures were subsequently used to construct ontology-level validation constraints.

The resulting ontology statistics are shown in Table II.

TABLE II
ONTOLOGY CONSTRUCTION STATISTICS

Dataset	Individuals	Obj. Prop.	Data Prop.	Assertions
Wikidata	1,974	7	1	1,600
DBpedia	1,851	7	0	1,600
FB15K-237	2,376	192	0	1,600
WN18RR	2,895	10	0	1,600
YAGO3-10	2,824	33	0	1,600

The ontology construction process was standardized to ensure that the validation results reflected the behavior of the HermiT-based approach rather than differences in implementation across datasets.

VI. EVALUATED ERROR CATEGORIES

Four error categories were evaluated in the experiment.

A. Domain Error

A Domain Error occurs when an entity is used as the subject of a relation in a way that violates the expected domain constraint of that relation.

The validation process compares the subject type implied by the KG assertion with the domain constraint represented in the ontology.

B. Range Error

A Range Error occurs when the object of a relation does not satisfy the expected range constraint.

The ontology represents relation range information, allowing HermiT to determine whether the corresponding assertions are semantically compatible.

C. Duplicate Fact

A Duplicate Fact represents a repeated assertion of an already existing fact.

Because OWL follows set-based semantics, exact duplicate axioms are naturally collapsed during ontology construction. Therefore, the experiment uses controlled duplicate-fact cases to evaluate the validation pipeline.

D. Missing Relation

Missing Relation validation evaluates whether a candidate relation is compatible with the available ontology constraints.

The experiment treats missing relations as candidate compatibility cases rather than assuming that every absent relation is false. This distinction is important because OWL follows the Open World Assumption.

VII. EXPERIMENTAL PROCEDURE

The validation experiment followed a common pipeline for all datasets:

- 1) Load the locally available KG dataset.
- 2) Select the controlled KG sample.
- 3) Construct the corresponding OWL ontology.
- 4) Define dataset-specific relation constraints.
- 5) Generate controlled validation cases.
- 6) Execute HermiT reasoning.
- 7) Record validation outcomes.
- 8) Calculate classification metrics.

The random seed was fixed to 42 wherever randomized sampling was required.

The evaluation pools were constructed separately for each applicable error category. The resulting cases were evaluated using the same metric definitions across all datasets.

VIII. EVALUATION METRICS

The experiment uses Accuracy, Precision, Recall, and F1-score.

Accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Precision is defined as:

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

The F1-score is defined as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

Although TP, TN, FP, and FN are used internally to calculate the metrics, they are not reported in the main results tables to maintain compact IEEE-format presentation.

IX. MEASURED RESULTS

The measured results for the four evaluated error categories are presented in Tables III–VII.

All decimal metric values are reported to three decimal places.

A. Wikidata

TABLE III
HERMIT VALIDATION RESULTS FOR WIKIDATA

Category	Cases	Acc.	Prec.	Rec.	F1
Domain	30	0.567	0.536	1.000	0.698
Range	30	0.567	0.536	1.000	0.698
Duplicate Fact	74	1.000	1.000	1.000	1.000
Missing Relation	30	0.533	0.519	0.933	0.667

For Wikidata, duplicate-fact validation achieved perfect performance. Domain and range validation achieved identical results, while missing-relation validation produced a lower F1-score.

B. DBpedia

TABLE IV
HERMIT VALIDATION RESULTS FOR DBPEDIA

Category	Cases	Acc.	Prec.	Rec.	F1
Domain	50	0.840	0.758	1.000	0.862
Range	50	0.840	0.758	1.000	0.862
Duplicate Fact	100	1.000	1.000	1.000	1.000
Missing Relation	50	0.680	0.680	0.680	0.680

DBpedia produced comparatively strong domain and range validation performance. Duplicate-fact validation again achieved perfect results, while missing-relation validation achieved balanced precision and recall.

C. FB15K-237

TABLE V
HERMIT VALIDATION RESULTS FOR FB15K-237

Category	Cases	Acc.	Prec.	Rec.	F1
Domain	38	0.526	0.514	1.000	0.679
Range	38	0.526	0.514	1.000	0.679
Duplicate Fact	100	1.000	1.000	1.000	1.000
Missing Relation	38	0.526	0.514	1.000	0.679

For FB15K-237, the measured domain, range, and missing-relation results were identical. Duplicate-fact validation again achieved perfect performance.

D. WN18RR

TABLE VI
HERMIT VALIDATION RESULTS FOR WN18RR

Category	Cases	Acc.	Prec.	Rec.	F1
Domain	8	0.500	0.500	1.000	0.667
Range	8	0.500	0.500	1.000	0.667
Duplicate Fact	100	1.000	1.000	1.000	1.000
Missing Relation	8	0.375	0.429	0.750	0.545

WN18RR produced perfect duplicate-fact validation performance. Domain and range validation produced an F1-score of 0.667, while missing-relation validation produced the lowest F1-score among the four evaluated categories.

E. YAGO3-10

TABLE VII
HERMIT VALIDATION RESULTS FOR YAGO3-10

Category	Cases	Acc.	Prec.	Rec.	F1
Domain	2	0.500	0.500	1.000	0.667
Range	2	0.500	0.500	1.000	0.667
Duplicate Fact	100	1.000	1.000	1.000	1.000
Missing Relation	2	0.500	0.500	1.000	0.667

YAGO3-10 also achieved perfect duplicate-fact validation performance. The domain, range, and missing-relation categories produced an F1-score of 0.667.

X. OVERALL RESULTS SUMMARY

The measured results demonstrate a clear difference between duplicate-fact validation and the other evaluated categories.

TABLE VIII
SUMMARY OF HERMIT VALIDATION PERFORMANCE

Dataset	Domain F1	Range F1	Duplicate F1	Missing F1
Wikidata	0.698	0.698	1.000	0.667
DBpedia	0.862	0.862	1.000	0.680
FB15K-237	0.679	0.679	1.000	0.679
WN18RR	0.667	0.667	1.000	0.545
YAGO3-10	0.667	0.667	1.000	0.667

Across all five datasets, duplicate-fact validation achieved an F1-score of 1.000. Domain and range validation produced similar results within each dataset because both rely on ontology-level relation constraints. Missing-relation validation showed greater variation across datasets.

XI. APPLICABILITY ANALYSIS

The applicability of ontology-based validation depends strongly on the semantic information available in the underlying KG.

Domain and range validation can be performed when relation-level structural constraints can be extracted or defined. These constraints allow the reasoner to identify assertions that violate the expected subject or object semantics.

Duplicate-fact validation is applicable across all five datasets because duplicate assertions can be represented through controlled validation cases.

Missing-relation validation is more difficult because an absent assertion does not necessarily indicate an error under the Open World Assumption. Therefore, the experiment evaluates missing relations as controlled candidate compatibility cases rather than treating every absent relation as invalid.

XII. DISCUSSION

The experimental results indicate that Hermit provides reliable ontology-level reasoning for validation tasks that can be expressed through explicit semantic constraints.

The consistently perfect duplicate-fact results are expected from the controlled representation of repeated assertions and the set-oriented nature of OWL axioms.

The lower performance observed for domain, range, and missing-relation cases demonstrates that reasoning alone does not guarantee perfect error detection. The quality and coverage of the constraints available from the underlying KG have a direct effect on validation performance.

The difference between datasets is therefore not solely a property of the HerMiT reasoner. It also reflects differences in relation structure, available semantic information, and the ability to construct meaningful ontology constraints.

XIII. LIMITATIONS

Several limitations should be considered when interpreting the results.

First, the ontology constraints are constructed from the available dataset information and therefore should be interpreted as experimental validation constraints rather than authoritative external ontologies.

Second, the controlled sample size is limited to maintain a common experimental scale across datasets.

Third, missing-relation validation cannot be interpreted as proof that an absent relation is factually false because OWL follows the Open World Assumption.

Fourth, the evaluated results depend on the quality and coverage of the extracted relation signatures.

Finally, the experiment evaluates ontology-based validation only and does not claim that HerMiT can detect every possible KG error.

XIV. REPRODUCIBILITY

The experiment was implemented using a shared validation script and separate notebooks for each dataset.

The experimental organization includes:

- hermit_validation.py
- create_notebooks.py
- run_experiments.py
- 01_Wikidata_HerMiT.ipynb
- 02_DBpedia_HerMiT.ipynb
- 03_FB15K-237_HerMiT.ipynb
- 04_WN18RR_HerMiT.ipynb
- 05_YAGO3-10_HerMiT.ipynb
- HerMiT_Master_Results.xlsx
- requirements.txt

A fixed random seed of 42 was used to improve reproducibility of controlled sampling and case generation.

XV. CONCLUSION

This study empirically evaluated HerMiT-based ontology validation across Wikidata, DBpedia, FB15K-237, WN18RR, and YAGO3-10. Four error categories were evaluated: Domain, Range, Duplicate Fact, and Missing Relation.

The results show that duplicate-fact validation achieved an F1-score of 1.000 across all five datasets. Domain and range validation showed dataset-dependent performance, while missing-relation validation exhibited comparatively greater variation.

The findings demonstrate that ontology-based reasoning can provide effective KG validation when appropriate semantic constraints are available. However, validation performance is influenced by the structural and semantic characteristics of the underlying KG. These results provide a reproducible baseline for comparing ontology-based validation with other KG validation approaches in the broader empirical evaluation.

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